

Identification of plasmids by PCR based replicon typing in bacteremic *Klebsiella pneumoniae*

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ABSTRACT

Background: *Klebsiella pneumoniae* is a notorious pathogen with plasmid mediated resistance to all classes of antibiotics. It is important to determine the plasmid profile coding for resistance genes. Plasmid profile varies among geographical regions and tracking the types helps in determining the MDR and XDR *K. pneumoniae* spread especially in hospital setting. Aim of the present study was to determine the plasmid profile and types among bacteraemic *K. pneumoniae*.

Materials and methods: Ninety consecutive *K. pneumoniae* collected over a period of three months from blood cultures were characterised by PCR for plasmid profile. Inc plasmid types were determined by PCR based replicon typing (PBRT) and carbapenemases were determined by multiplex PCR. For a subset of isolates hybrid assemblies were developed by sequencing with Ion Torrent and MinIon.

Results: Overall, PBRT showed 29% of isolates carried four plasmids including IncHI1B, IncFIA, IncFII(K) and IncR. The most common type of plasmid was IncHI1B (93%) followed by IncFII_K (89%) and IncR (82%). IncFIA was predominant among carbapenem resistant isolates. Almost all plasmids identified in *K. pneumoniae* were AMR plasmids, except two isolates which had virulence plasmids. IncX3 plasmid observed in this study was previously reported to be self-disseminating. Furthermore, the hybrid genome sequencing revealed complete structural arrangements of plasmids, which would be missed in short-read sequencing. NDM and OXA48-like were co-produced in 59% of the carbapenem resistant isolates. *bla*_{OXA-232} was present on ColKP3; *aac*(6')-Ib3 and *rmtF* on IncFIB.

Conclusion: Diverse plasmid profile among the successive *K. pneumoniae* isolates indicates the transfer of resistance genes through different types of plasmids. IncHI1B, IncFIA, IncFII_K and IncR were the prevalent plasmid types. Hybrid assembly revealed *bla*_{OXA-232} was present on ColKP3 unlike global reports of IncL/M. Hybrid assemblies provide better plasmid structure that long and short read assemblies. There was no significant association of β -lactamases with specific Inc groups in this study.

1. Introduction

World Health Organisation (WHO) has recently classified ESBL (extended spectrum β -lactamase) producing and carbapenem resistant *Enterobacteriaceae* as a priority 1, critical pathogen, in the list for developing new antimicrobial agents. ESBL and carbapenem resistance among *Enterobacteriaceae* is mainly plasmid encoded belonging to various plasmid families. Recently, plasmid encoded colistin resistance has been emerging across human, animal and environmental sources. It is thus clear that plasmids play a prime role in antimicrobial resistance

and infection dissemination.

Plasmid types were described based on the stability of plasmids during conjugation and was called plasmid incompatibility (Inc) which is defined as the inability of two plasmids belonging to the same Inc group to be propagated stably in the same cell [1,2]. There are 27 Inc groups recognized in *Enterobacteriaceae* by the Plasmid Section of the National Collection of Type Culture (Colindale London, UK). PCR-Based Replicon Typing (PBRT) scheme has been developed for plasmids in *Enterobacteriaceae* which is based on *repA* gene [3].

Aim of the present study was to determine the plasmid profiles and

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types among *K. pneumoniae* isolated from bacteremia. Plasmid profile of *K. pneumoniae* was also determined by whole genome shotgun sequencing followed by hybrid genome assembly to study the trend of multi-drug resistance, and association of antimicrobial resistance (AMR) and virulence genes. Plasmid profile varies among geographical regions and tracking of the types helps in determining the multi-drug resistant (MDR) and extensively drug resistant (XDR) *K. pneumoniae* spread especially in hospital setting.

2. Materials and methods

2.1. Isolates and phenotypic characterisation

A total of 90 consecutive *K. pneumoniae* isolated from blood cultures from January to March 2015 at the department of Clinical Microbiology, Christian Medical College, Vellore, India, were included in the study. The BacT/ALERT (Biomérieux) system was used for incubation of blood cultures. The isolates were identified using standard biochemical reactions [4]. Isolate from the first blood culture of each patient was included in the study.

Antimicrobial susceptibility testing was performed by Kirby Bauer disc diffusion as recommended by CLSI [5] for different classes of antimicrobials including cefotaxime (30 µg), ceftazidime (30 µg), piperacillin/tazobactam (100/10 µg), imipenem (10 µg), meropenem (10 µg), ciprofloxacin (5 µg), gentamicin (10 µg), amikacin (30 µg) and tigecycline (15 µg). The results were interpreted according to CLSI 2015 guidelines except for tigecycline which was interpreted according to FDA breakpoints [5,6]. *E. coli* ATCC 25922 and *E. faecium* ATCC 29212 were used as control for susceptibility testing. The isolates were classified into three groups based on their susceptibility patterns: (a) isolates that are susceptible to cephalosporins; (b) extended spectrum β-lactamase (ESBL) producers and (c) carbapenem resistant isolates. The isolates were classified as ESBL producers if the zone diameter of ceftazidime and cefotaxime was ≤22 mm and ≤27 mm respectively and an increase in zone diameter of ≥5 mm was observed when tested with clavulanic acid. They were classified as carbapenem resistant if the zone diameter for meropenem was ≤19 mm. Further, the isolates that were resistant to meropenem were subjected to E-test for meropenem (Biomérieux) for determining minimum inhibitory concentration (MIC).

2.2. Genotypic characterisation

Bacterial DNA was extracted from 18 to 24 h cultures by Qiasymphony (Qiagen, Hilden, Germany) as per manufacturer's instructions. Multiplex PCR was performed for detection of genes encoding beta-lactam resistance which includes EBSL, CTX-M, ampC and carbapenemases. Multiplex PCR for EBSL genes included *bla_{SHV}*, *bla_{TEM}*, *bla_{PER}*, *bla_{VEB}* and *bla_{GES}* and was performed as previously described by Dallene et al., 2010 [7]. Multiplex PCR for CTX-M genes including *bla_{CTX-M-1}*, *bla_{CTX-M-2}*, *bla_{CTX-M-8}*, *bla_{CTX-M-9}* and *bla_{CTX-M-25}* was performed according to Woodford et al., 2004 [8]. Multiplex PCR for AmpC β-lactamases including *bla_{ACT}*, *bla_{ACC}*, *bla_{CIT}*, *bla_{DHA}* and *bla_{FOX}* was performed as previously described [9]. Carbapenemase genes detected by multiplex PCR included *bla_{NDM}*, *bla_{VIM}*, *bla_{IMP}*, *bla_{OXA 48-like}*, *bla_{SPM}* and *bla_{KPC}* and was performed as previously described [10–12].

2.3. Plasmid typing

PCR based replicon typing to classify plasmids into various incompatible (Inc) groups was performed as described by Carattoli et al., 2005 and Villa et al., 2010 [3,13]. PCR was performed for determination of the following plasmid types: IncFII, IncFII_K, IncFIA, IncFIB, IncHI1B, IncN, IncA/C, IncL/M and IncB/O. Positive controls for various plasmid types were obtained from whole genome sequencing of *K. pneumoniae* (NEWA00000000, PKOM00000000, NQYU00000000) and *Shigella* spp. (PDXU00000000, NMYB00000000, MDJK00000000)

from this study collection. Gel electrophoresis with 2% agarose was performed to visualise the PCR amplicons.

2.4. Whole genome sequencing

Seven isolates were subjected to whole genome sequencing using Ion Torrent PGM platform with 400bp chemistry (long reads). Raw reads were assembled using Assembler SPAdes v.5.0 software in Torrent suite server version 4.4.3. Hybrid genome assembly by Ion Torrent + ONT MinION reads was performed for these seven isolates [14]. Sequencing was done as per the standard protocols. The Fast 5 file was generated from minion sequencing and the reads were base-called with Albacore 2.0.1. Furthermore, the adapters were trimmed off using Porechop (<https://github.com/rrwick/Porechop>). Canu 1.7 was used for Minion error correction with genome size of 5.0 MB as input. Minimap2 (version 2.1) and Miniasm (V 0.2) were used to assemble the corrected reads and subsequently the assembly was polished with multiple rounds of Racon. The sequences were deposited in the NCBI and accession numbers were obtained.

The genomes were annotated using RAST (Rapid Annotation using Subsystems Technology- <http://rast.nmpdr.org/>), Patric (Pathosystems Resource Integration Centre - <https://www.patricbrc.org/>) and the National Centre for Biotechnology Information Prokaryotic Genomes Automatic Annotation Pipeline (PGAAP) softwares. The resistance genes were identified using ResFinder version 2.1 (<https://cge.cbs.dtu.dk/services/ResFinder/>) and Multi-locus Sequence typing (MLST) was determined using database at <https://cge.cbs.dtu.dk/services/MLST/>. Plasmids present in the genome were identified by PlasmidFinder version 1.3 available at <https://cge.cbs.dtu.dk/services/PlasmidFinder/>.

2.5. Clinical details

The demographic and clinical details of patients were obtained from electronic medical records. The source of infection, site of acquisition and co-morbid conditions were considered for the study. The site of acquisition was classified as mentioned below:

Hospital acquired infection: blood culture positive more than 48 h after admission to hospital.

Healthcare associated infection: blood culture positive in the first 48 h of admission in a patient hospitalized for at least two days within the last 90 days, receiving specific home nursing care or attending hospital or haemodialysis clinic in the 30 days before infection.

Community acquired infection: blood culture positive within 48 h of admission in a patient without prior hospitalization in last 60 days.

3. Results

3.1. Antimicrobial susceptibility of *K. pneumoniae*

Based on the antimicrobial susceptibility testing, the 90 isolates included in the study were assigned into three groups: first group consisted of the isolates that were susceptible to cephalosporins included 32 isolates (35.5%), second group comprising of ESBL producers consisted of 24 isolates (26.6%) and the third group with carbapenemase producers consisted of 34 isolates (37.7%). The MIC of meropenem by E-test ranged from 0.38 µg/ml to >32 µg/ml among the isolates that were carbapenem resistant by disc diffusion. Susceptibility of the study isolates to various antimicrobials tested in mentioned in Fig. 1.

3.2. Molecular mechanisms of β-lactam resistance

For the isolates that were susceptible to all antimicrobials, PCR for detection of *bla_{SHV}* and *bla_{TEM}* was performed. Among these isolates, all produced SHV since *bla_{SHV}* is intrinsic to *K. pneumoniae*. Five isolates produced SHV and TEM which were not detected as ESBL producers by

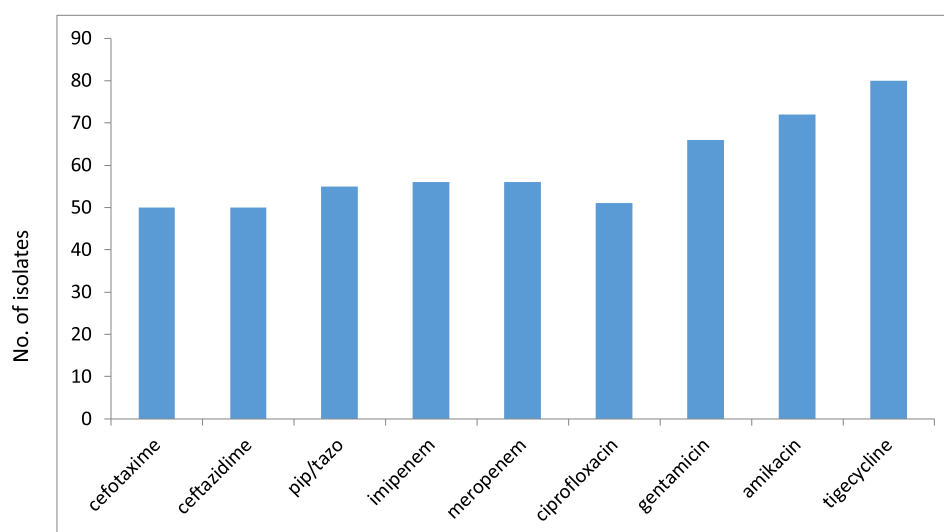


Fig. 1. Susceptibility of *K. pneumoniae* to various antimicrobials as tested by disc diffusion.

disc diffusion. Results of β -lactamases present among the isolates that were phenotypically classified as ESBL and carbapenem resistant isolates are shown in Table 1.

3.3. Prevalent plasmid profiles and types

PBRT revealed multiple replicon types in each isolate. Among these, the most common plasmid profiles observed were IncHI1B, IncFIA, IncFII_K and IncR, together being present in 29% followed by combination of other plasmid types in 10%. The most common plasmid type was IncHI1B (93%) followed by IncFII_K (89%) and IncR (82%). Results of plasmid typing are shown in Table 2. The distribution of IncHI1B, IncR and IncFII_K were comparable among the three groups of *K. pneumoniae*. However, IncFIA was frequently found among the carbapenem resistant isolates (70%). IncN and IncB/O were absent among the study isolates.

Table 1

Distribution of molecular mechanisms of β -lactam resistance based on antimicrobial susceptibility of *K. pneumoniae*.

	ESBL producers (n = 24)	Carbapenem resistant (n = 34)
ESBL enzymes		
• SHV	3 (13%)	1 (3%)
• TEM	0	0
• SHV and TEM	0	1 (3%)
• SHV and CTX-M-1	1 (4%)	4
• SHV, TEM and CTX-M-1	17 (71%)	25 (76%)
• TEM and CTX-M-1	1 (4%)	2 (6%)
• Negative	1 (4%)	0
AMP C enzymes		
• ACC	1 (4%)	0
• CIT	1 (4%)	4 (12%)
• Negative	22 (92%)	29 (88%)
Carbapenem resistant enzymes		
• NDM	3 (12.5%)	5 (15%)
• OXA48 like	0	3 (9%)
• NDM and OXA48 like	1 (4%)	20 (59%)
• VIM	1 (4%)	0
• NDM, VIM	0	1 (3%)
• Negative	19 (79.5%)	4 (12%)
• IMP	0	0
• SPM	0	0
• KPC	0	0

3.4. Association of nature of infection with AMR and plasmid types

Among the 90 cases included, 60 cases (67%) were due to hospital acquired *K. pneumoniae*, 17 (19%) were due to healthcare associated infection and 13 (14%) were due to community acquired infections. The distribution of plasmids among community acquired and healthcare associated infections is mentioned in Table 2.

Overall, 44% (40) of the isolates carried three plasmids followed by 29% (26) with four plasmids each. Among the community acquired *K. pneumoniae*, 54% (7) carried three plasmids while among the healthcare and hospital acquired *K. pneumoniae*, 40% (36) of the isolates carried five plasmids. Among the cephalosporin susceptible isolates, 38% (12) and 34% (11) carried three and four plasmids respectively. Among the ESBL *K. pneumoniae* and carbapenem resistant *K. pneumoniae*, 55% (13) and 42% (14) of the isolates respectively carried three plasmids.

Interestingly, none of the community acquired *K. pneumoniae* was carbapenem resistant and 69% (9) were cephalosporin susceptible and 31% (4) were ESBL *K. pneumoniae*. Among the healthcare and hospital acquired *K. pneumoniae* (77), 43% (33) were carbapenem resistant followed by 30% (23) of ESBL producers and 26% (20) of cephalosporin susceptible isolates.

3.5. Hybrid genome assembly

From the hybrid assembly results, the AMR genes on each plasmid were identified and are mentioned in Table 3. Interestingly, *bla*_{OXA-232} was present on ColKP3 and none of the *bla*_{OXA-232} carried IncL/M plasmid. Genes encoding aminoglycoside resistance were present on diverse plasmid types as shown in Table 3 except for *rmtF* which was present on IncFIB plasmid. Similarly, genes coding for resistance to other classes of antimicrobials such as β -lactams, quinolones and sulphonamides were distributed among diverse plasmid types. Some plasmids such as Col440I, Col440II and ColRNAI did not carry any resistance or virulence genes. These Col plasmids are a part of the bacterial virulence since they code for colicins. In two isolates; BA4656 and BA34918, virulence plasmid was identified which also carried the *catA1* gene coding for chloramphenicol resistance.

Fig. 2a shows the comparison of IncFIB plasmid among four study isolates as constructed using hybrid assembly. Among the four isolates, IncFIB carried antimicrobial resistance (AMR) genes such as *rmtF*, *aac* (6')-Ib3, *qnrB1* and *ARR-2*. However, some AMR genes including *bla*_{TEM}, *bla*_{CTX-M-15}, *mph* and others, were lost from the IncFIB (pQil) and were found on other plasmids in these isolates. Interestingly, two isolates

Table 2Results of replicon based plasmid typing for *K. pneumoniae*.

Plasmids (n = 90)	No. of positive isolates (%)	Susceptible isolates (n = 32)	ESBL isolates (n = 24)	Carbapenem resistant isolates (n = 34)	Community acquired infection (n = 13)	Hospital acquired and healthcare associated infection (n = 77)
IncL/M	18 (20%)	8 (25%)	4 (17%)	6 (18%)	3 (23%)	15 (19%)
IncR	74 (82%)	28 (88%)	21 (88%)	24 (71%)	12 (92%)	62 (81%)
IncA/C	6 (7%)	1 (3%)	1 (4%)	4 (12%)	1 (8%)	5 (6%)
IncFIA	38 (42%)	10 (31%)	5 (21%)	23 (68%)	4 (31%)	34 (44%)
IncFII	7 (8%)	3 (9%)	1 (4%)	3 (9%)	1 (8%)	6 (8%)
IncFII _K	80 (89%)	26 (81%)	23 (96%)	30 (88%)	13 (100%)	67 (87%)
IncHI1B	84 (93%)	28 (88%)	21 (88%)	30 (88%)	12 (92%)	72 (94%)
IncFIB	7 (8%)	3 (9%)	2 (8%)	2 (6%)	2 (15%)	5 (6%)

BA4656 and BA34918 have identical IncFIB (pQil) plasmid sequence. However, the two other isolates BA28434 and BA2172 share similar sequences which differ from that of IncFIB of BA4656. The latter two isolates lack certain regions coding for *tra* operon, type 1 restriction modification system, Domain of Unknown Function (DUF) proteins and some insertion sequences. BA2172 was isolated during 2016 while BA28434 during 2017. These two plasmids have not undergone changes while the IncFIB isolated during 2018 have acquired additional mobile genetic elements such as insertion sequences of various families. Also, the *tra* operon, which enable transfer of genetic elements are present among the plasmids isolated during 2018.

Fig. 2b shows comparison of IncFIB_K among three isolates. BP327 and BA1559 have identical IncFIB_K and were isolated 8 days apart although from different wards. BA33875 was isolated during 2016 and has a several differences from the other two IncFIB_K. This plasmid also had diverse AMR genes (Table 3). This plasmid also coded for heavy metal resistance to agents such as copper, lead and arsenic in all three isolates while resistance to mercury was coded for only in IncFIB_K of BA33875. Hence, the same plasmid type isolated over different time frame shows several differences.

4. Discussion

Among the *K. pneumoniae* isolates tested, cephalosporin susceptible, ESBL and carbapenem resistant isolates accounted for 35.5%, 26.6% and 37.7% respectively. The percentage of ESBL and carbapenem resistant *K. pneumoniae* in the study centre is about 61% and 32% respectively which is comparable to other reports from India [15]. Another study done in India on *K. pneumoniae*, also reports IncFIIK as the commonest plasmid type followed by IncR [16]. This study also included non-ESBL, ESBL and carbapenem resistant *K. pneumoniae* similar to the present study. IncHI1B was found to be about 14% by Nithya et al. [16], unlike the present study wherein IncHI1B was one of the commonest plasmids found in 93% of the isolates. IncX2, IncK and IncA/C reported by Nithya et al., were absent in the present study [16]. Khajuria et al., reported *bla*_{NDM-1} on IncA/C plasmid and *bla*_{CTX-M-15} on IncFIA and IncFIB plasmids [17]. Subramanian et al. from Southern India report IncHI3, IncFIIK, IncFII1B among the NDM-1 producing *K. pneumoniae* [18]. However, the present study lacked IncHI3 type of plasmids. Though the studies have been conducted from Southern India, the plasmid profile appears to be diverse and distinct in each hospital. This indicates the rapid emergence and spread of *K. pneumoniae* with different plasmid types and hence the antimicrobial resistance genes can be transferred by diverse plasmid types. Since the plasmids are compatible among all the members of *Enterobacteriaceae*, there can be horizontal transmission and rapid spread of resistance to organisms such as *E. coli* which have low resistance rates.

IncI1-Iy and IncFIA-FIB were commonly seen among North Indian ESBL producing *K. pneumoniae* [19]. A recent study reports the presence of ESBL genes predominantly among IncF and IncI plasmids [20]. Recently there has been a shift from *bla*_{NDM} to *bla*_{OXA48-like} in most of the carbapenem resistant *K. pneumoniae* in the present study centre [15]. From the whole genome sequencing of the selective study isolates,

*bla*_{OXA-232} was found to be carried by ColKP3. Similar results have been reported by Simner et al., and Shu et al., where *bla*_{OXA-232} and *bla*_{OXA-181} were carried on ColKP3 [21,22]. IncA/C, IncR and IncFIIK were predominant among carbapenem resistant *K. pneumoniae* which mainly contained *bla*_{OXA48-like} in Turkey [23].

In a previous report at the study centre, the plasmid profile among ESKAPE pathogens was determined [24]. Among *E. coli* and *K. pneumoniae*, 56.5% were IncF group of plasmids among which IncFIA and IncFII were predominant. Col group of plasmids comprised 18% among these two organisms. IncX and IncH groups were present among ≤5% of the study isolates. However, in the present study IncHI1B was the most predominant plasmid type seen among *K. pneumoniae*. Also, IncFIIK and ColKP3 were frequently found among *K. pneumoniae* similar to the present study. Another report from the study centre showed the presence of *bla*_{NDM-7} on IncX3 plasmid in *E. coli*. However, among *K. pneumoniae*, IncX3 characterised using hybrid assembly did not contain *bla*_{NDM} gene [25].

Hybrid assemblies have the advantage of the obtaining complete genome and hence give a better insight into plasmid structure when compared to IonTorrent which gives fragmented reads and hence completeness of information is lacking [14]. MinION gives long reads but with high error rates and hence the present study used hybrid assembly to obtain complete plasmid structure in order to determine the plasmid profiles. In this study, hybrid assembly revealed the complete structural arrangements of AMR genes in various plasmids. This also add the information on the multiple copies of AMR genes in each plasmid which may directly relate to the rate of expression. Accordingly, in one isolate carrying ColKP3, two copies of *bla*_{OXA-232}, and in other isolates with IncHI1B and IncA/C2 two copies of *bla*_{CTX-M-15} were found. A study done on *K. pneumoniae* from various regions of South east Asia, used hybrid assembly obtained by combining reads of Illumina and MinION, have identified novel virulence plasmids [26]. This method enables complete plasmid characterisation and helps in obtaining newer insights.

Among *K. pneumoniae*, there has always been a concern about virulence and resistance genes being disseminated by a single plasmid but there were no such reports until the recent report by Lam and colleagues. This plasmid was a hybrid containing regions from IncHI1B, IncFIIK and IncFIB_K [30]. In the present study, *catA1* was present in two isolates on virulence plasmids which carried *rmpA/rmpA2* and genes coding for heavy metal resistance.

Mobilization potential of plasmids depends on the replicating machinery that it carries. Usually, smaller plasmids use the host replicative enzymes to multiply, whereas larger plasmids carry genes required for their own replication. The mobilization of the plasmid affects the dissemination potential of the AMR genes as the host-range of multiple plasmid types differ. For instance, *bla*_{CTX-M-15} in IncFII plasmid is not expected to disseminate to *Acinetobacter* or *Pseudomonas* spp., since the host range of IncFII plasmid is limited to *Enterobacteriaceae*. IncP, IncA/C, or IncQ plasmids were known to have a larger host range including *Pseudomonas* spp. and gram-positive bacteria [27]. IncF plasmids need DNA gyrase, DnaB, DnaC, DnaG, SSB, and DNA polymerase III proteins for their replication, while IncQ plasmids are independent of DnaA,

Table 3Distribution of antimicrobial resistance genes on plasmids in *K. pneumoniae* inferred from hybrid assemblies of whole genome sequences.

Plasmid type	BA33875	BA4656	BA34918	BP327	BA1559	BA2172	BA28434
	ST101	ST23	ST23	ST15	ST15	ST231	ST231
IncFIB(pQil)	absent	<i>aac(6')-lb3</i> , <i>rmtF</i> , <i>qnrB1</i> , <i>ARR-2</i>	<i>rmtF</i> , <i>aac(6')-lb3</i> , <i>qnrB1</i> , <i>ARR-2</i>	absent	absent	<i>rmtF</i> , <i>aac(6')-lb3</i> , <i>bla_{CTX-M-15}</i> , <i>bla_{TEM-1B}</i> , <i>qnrS1</i> , <i>catA1</i> , <i>ARR2</i>	<i>aadA2</i> , <i>rmtF</i> , <i>aac(6')-lb3</i> , <i>bla_{CTX-M-186}</i> , <i>aac(6')-lb-cr</i> , <i>mphA</i> , <i>catA1</i> , <i>ARR2</i> , <i>sul1</i> , <i>dfrA12</i>
IncFIBK	<i>armA</i> , <i>aadA1</i> , <i>aadA2</i> , <i>aac(6')-lb</i> , <i>sul1</i> , <i>msrE</i> , <i>mphE</i> , <i>dfrA12</i> , <i>bla_{SHV-33}</i> , <i>bla_{OXA-9}</i>	Absent	absent	<i>aph(3')-la</i> , <i>aac(6')-lb-cr</i> , <i>bla_{CTX-M-15}</i> (2 copies), <i>bla_{OXA-1}</i> , <i>mphA</i> , <i>tetA</i>	<i>aadA2</i> , <i>aac(6')-lb-cr</i> , <i>aph(3')-la</i> , <i>bla_{TEM-1B}</i> , <i>bla_{OXA-1}</i> , <i>bla_{CTX-M-15}</i> , <i>mphA</i> , <i>sul1</i> , <i>dfrA12</i> , <i>tetA</i>	absent	absent
ColKP3	absent	Absent	<i>bla_{OXA-232}</i>	absent	absent	<i>bla_{OXA-232}</i>	<i>bla_{OXA-232}</i> (2 copies)
IncFIA	absent	Absent	absent	absent	absent	absent	<i>aadA2</i> , <i>ermB</i> , <i>sul1</i> , <i>dfrA12</i>
IncFII _K	<i>bla_{KPC-2}</i>	<i>aac(6')-lb-cr</i>	absent	absent	absent	absent	absent
IncHI1B	absent	Absent	absent	<i>aadA2</i> , <i>armA</i> , <i>aph(6)-ld</i> , <i>aph(3')-lb</i> , <i>bla_{CTX-M-15}</i> (2 copies), <i>bla_{TEM-1B}</i> , <i>mphE</i> , <i>msrE</i> , <i>sul1</i> , <i>sul2</i> , <i>dfrA12</i>	absent	absent	absent
IncA/C2	absent	<i>aph(6)-ld</i> , <i>aph(3')-lb</i> , <i>bla_{TEM-1B}</i> , <i>bla_{CMY-4}</i> , <i>bla_{CTX-M-15}</i> , <i>sul2</i> , <i>dfrA14</i>	<i>aph(6)-ld</i> , <i>aph(3')-lb</i> , <i>bla_{CTX-M-15}</i> (2 copies), <i>bla_{TEM-1B}</i> , <i>bla_{CMY-4}</i> , <i>sul2</i> , <i>dfrA14</i>	absent	absent	absent	absent
IncR	<i>bla_{SHV-13}</i> , <i>bla_{TEM-1B}</i> , <i>bla_{OXA-9}</i> , <i>aadA1</i> , <i>dfrA14</i> , <i>qnrB1</i> , <i>aac(3)-lld</i> , <i>aac(6')-lb</i>	Absent	absent	absent	absent	absent	absent
IncX3	absent	No AMR genes	<i>aac(6')-lb-cr</i> , <i>armA</i> , <i>msrE</i> , <i>mphE</i> , <i>sul1</i>	absent	absent	absent	absent
Col440I	absent	Absent	absent	No AMR genes	No AMR genes	No AMR genes	absent
Col440II	absent	No AMR genes	No AMR genes	No AMR genes	absent	absent	absent
ColRNAI	absent	No AMR genes	No AMR genes	No AMR genes	absent	absent	absent
IncFIB (pKPHS1)	absent	Absent	No AMR genes	No AMR genes	absent	absent	absent

Note: 'absent' refers to plasmid being absent in the isolates. 'No AMR genes' indicates the presence of plasmid but lack of AMR genes.

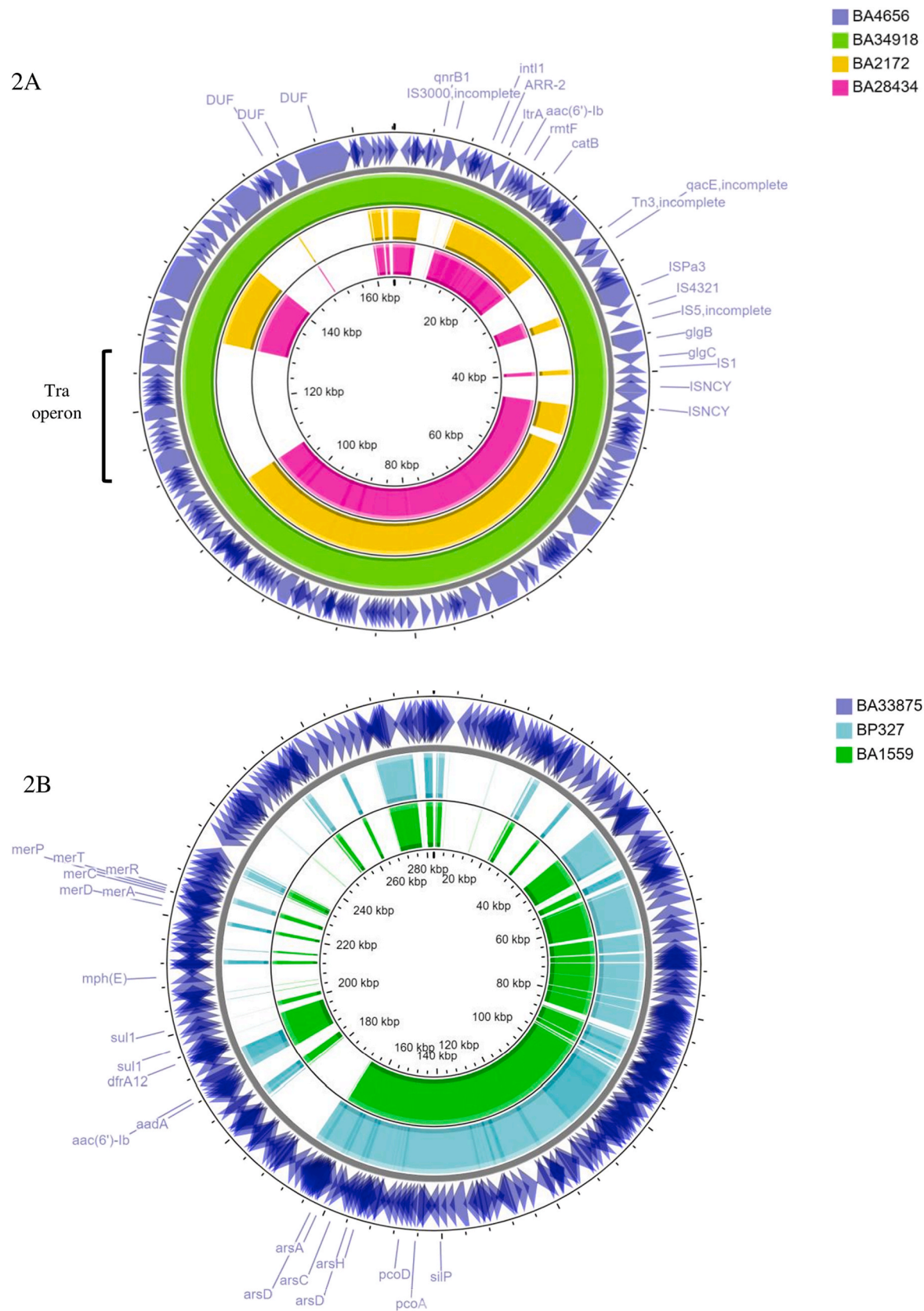


Fig. 2. a: Comparison of IncFIB (pQil) plasmid from four *K. pneumoniae* isolates; Fig. 2b: Comparison of IncFIB_K plasmid from three *K. pneumoniae* isolates.

DnaB, DnaC, and DnaG proteins, and they provide the proteins essential for their replication (RepA, RepB, and RepC) in almost every bacterial cell. However, they require expression of necessary plasmid-encoded proteins in the bacterial host and interaction between plasmid and host-proteins at specific stage during host cell cycle [27,28].

In addition, study by Psychogiou et al. had revealed the transfer of *bla*_{VIM-1} via self-transmissible IncN plasmids in *K. pneumoniae* [29]. However, in the present study IncN plasmids were absent. ColE or RSF1010 (IncQ) plasmid derivatives are not self-transmissible by conjugation and can be mobilized at a high frequency in the presence of a helper plasmid. Other classes of Col plasmids seen in the present study includes ColKP3, ColRNAI and Col440I which also require the host for transmission.

5. Conclusions

Diverse plasmid profile among the successive *K. pneumoniae* isolates indicates the transfer of resistance genes through different types of plasmids. IncHI1B, IncFIA, IncFIIK and IncR was the prevalent plasmid profile with IncHI1B, IncFIIK and IncR being the common plasmid types irrespective of the antibiogram among *K. pneumoniae* from bacteremia. Significantly increased incidence of IncFIA was seen among the carbapenem resistant isolates. Hybrid assembly revealed the co-existence of AMR and virulence determinants in a single plasmid. *Bla*_{OXA-232} was present on ColKP3 while *aac*(6')-*lb3* and *rmtF* were present on IncFIB plasmid. Other resistance genes were associated with different plasmid types. There was no significant association of other β -lactamases with specific Inc groups in this study. IncX3 plasmid observed in this study was previously reported to be self-disseminating. It is imperative to continuously monitor the plasmid profile to detect the AMR dissemination in the healthcare setting.

Author statement

Chaitra Shankar: Laboratory methods such as DNA extraction, PCR, whole genome sequencing, data analysis, manuscript writing and manuscript correction. Dhiviya Prabaa Muthurilandi Sethuvel: Laboratory methods such as DNA extraction and PCR. Ayyan Raj Neeravi: Laboratory methods such as DNA extraction, PCR, whole genome sequencing. Manigandan Venkatesan: Laboratory methods such as DNA extraction, PCR, whole genome sequencing. Naveen Kumar Devanga Ragupathi: Hybrid assembly and manuscript writing. Shalini Anandan: Study design, manuscript writing and correction. Balaji Veeraraghavan: Study design, manuscript correction. This research work is a part of the doctoral thesis of Ms. Chaitra Shankar under the Tamil Nadu Dr. M.G.R. Medical University, Chennai, Tamil Nadu, India.

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Declaration of competing interest

On behalf of all the authors, I, Dr. Balaji Veeraraghavan, declare that we do not have any conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.micpath.2020.104429>.

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